

Question Paper

Exam Date & Time: 23-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Basic Electrical Technology [ELE-1071]

Marks: 50

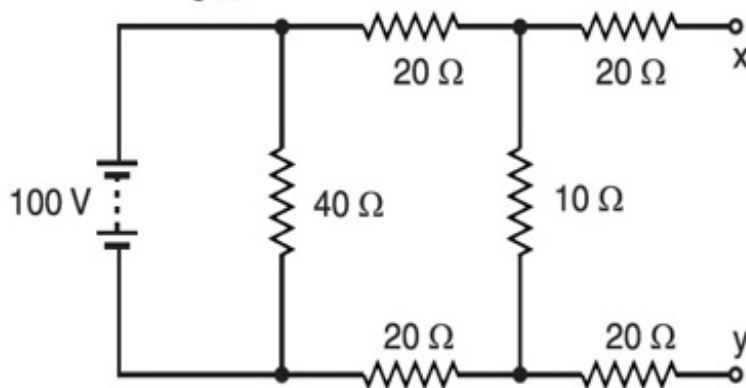
Duration: 180 mins.

PART-A

Answer all the questions.

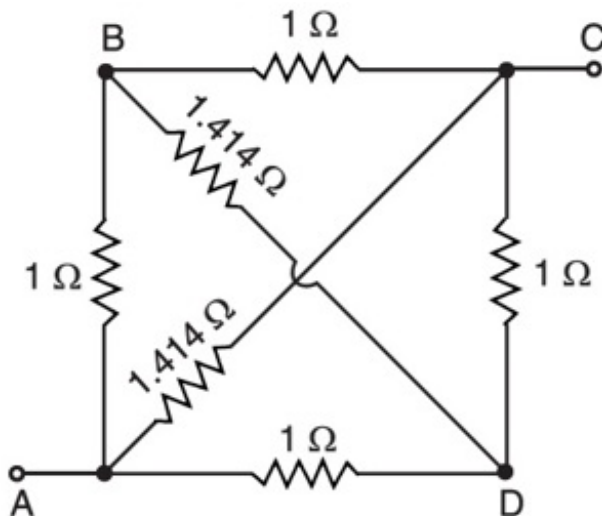
- 1) Calculate the power which would be dissipated in a $50\ \Omega$ resistor connected across **xy** in the network shown in Figure below. (5)

A)



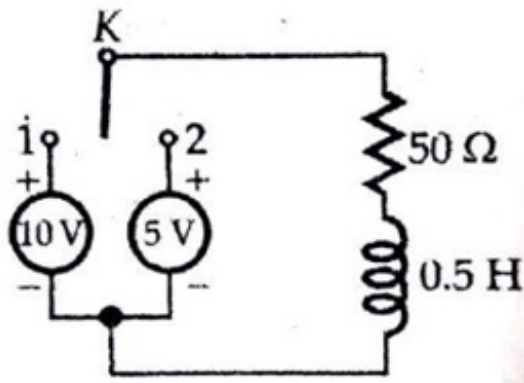
Course Outcome: "CO1" / Cognitive Level: "Application"

- B) A square and its diagonals are made of a uniform covered wire. The resistance of each side is $1\ \Omega$ and that of each diagonal is $1.414\ \Omega$ as shown in the figure below. Determine the resistance between two opposite corners of the square. (5)



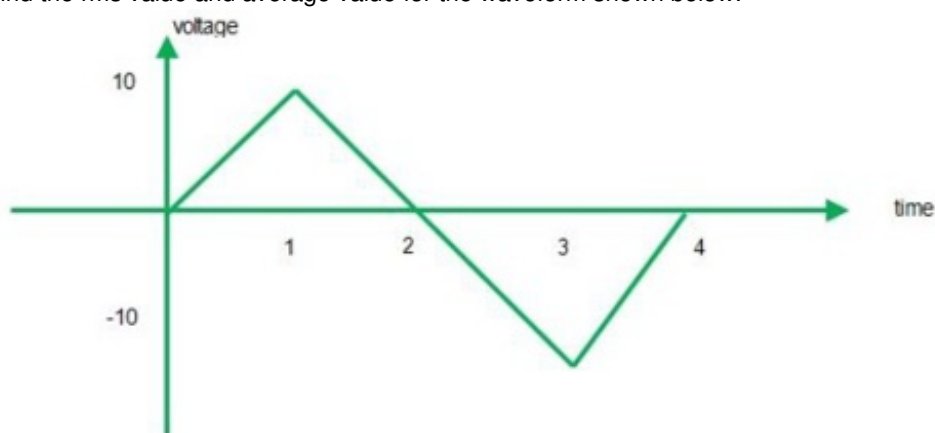
Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) In the circuit shown below, the switch is closed to position 1 at time $t = 0$ secs. Then at time $t = 0.5$ secs, the switch is moved to position 2. Find the expressions for the current through the circuit from 0 to 0.5 msecs and beyond 0.5 msecs. (4)
- A)



Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Find the rms value and average value for the waveform shown below. (3)

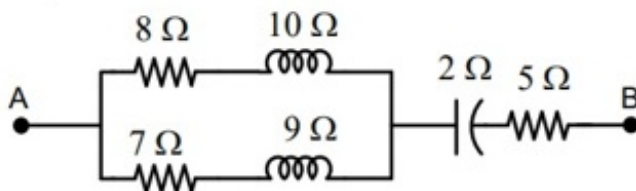


Course Outcome: "CO2" / Cognitive Level: "Analysis"

- C) A 120V, 60W lamp is to be operated on 220V, 50 Hz supply mains. In order that lamp should operate on correct voltage rating, calculate the value of (3)
- (i) Non-inductive resistance
(ii) Pure inductance

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) In the arrangement shown in the figure below. Calculate the impedance between AB and the phase angle between voltage and current. Also, calculate the total power consumed, if the applied voltage between AB is $200\angle 30^\circ$. (5)
- A)



Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Three identical coils having resistance of 10Ω and an inductance of 0.05 H each are connected in star across a 3 ϕ , 400 V, 50 Hz balanced supply. Calculate the line current and the power consumed. What will be the reading of two wattmeters connected to measure the total power. (5)

Course Outcome: "CO3" / Cognitive Level: "Application"

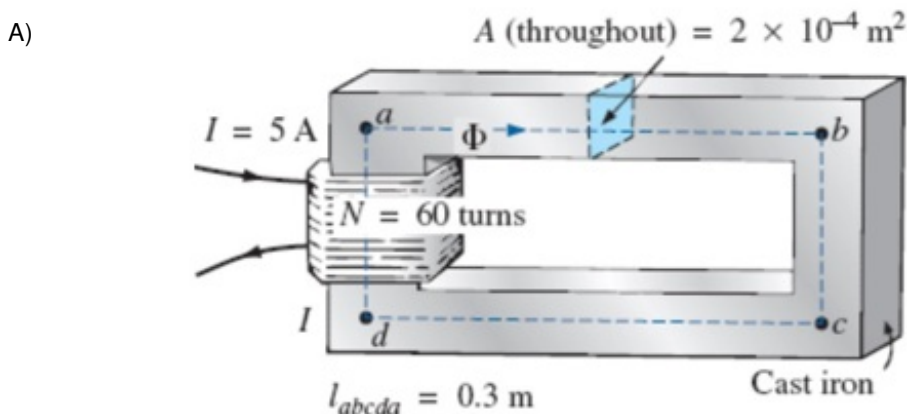
- 4) A 3-phase motor takes 5 kW at 0.8 p.f lagging from a 3-phase 400 V supply. If the motor is represented by a balanced delta- connected set of impedances, find the impedance in each phase. (5)
- A) What is the reactive power taken by the motor?

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) An iron ring having cross sectional area of 400 mm² and mean circumference of 500 mm carries a coil of 250 turns wound uniformly around it. Calculate (5)
- (a) the reluctance of the ring
- (b) current required to produce the flux of 1000 μwb in the ring. Relative permeability of iron is 400.

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Calculate the magnetic flux Φ for the magnetic circuit shown in the figure below. (3)



Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Explain the construction of synchronous motor. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- C) Explain the working of three phase digital energy meter. (4)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ELE-1071:Basic Electrical Technology

#	Course Outcome	Description	Marks
1	1	CO 1	14
2	2	CO 2	11
3	3	CO 3	10
4	4	CO 4	8
5	5	CO 5	7
Total			50

* NA - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ELE-1071:Basic Electrical Technology

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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	7(14%)
3	Application	7	31(62%)
4	Analysis	3	12(24%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	12	50 (100%)